

Chan Yat Kwan

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EDUCATION

The University of Hong Kong

Bachelor of Engineering in Computer Science | CGPA: 3.60

June 2026

Relevant Coursework: Data Structures and Algorithms, Database, Machine Learning, Cyber Security, OOP, Game Development, Mobile App Development, World Wide Web, Communication Network, Computer Organization, Software Engineering, Data Science

WORK EXPERIENCE

XR + AI Developer Intern | **VOTANIC Limited** | *Unity, C#, Docker, Node.js, Express.js*

June 2025 - August 2025

Develop XR and AI solution with XR platform integration

- Developed demo project in **Unity** in a team of 2, scripted in **C#**, utilized **machine learning** framework for **face detection**, to implement **off-axis projection** via user head tracking.
- Integrated a **photo-to-3D model generation** pipeline into a web-based 3D model editing application, using **Docker**, **Node.js**, and **Express.js** for containerization and deployment.

STEM Intern | **Up Innovations Limited** | *Unity, C#, AWS, Firebase*

July 2024 - August 2024

Develop interactive applications for Special Educational Needs

- Designed and built a game prototype in **Unity**, scripted in **C#**, utilizing **machine learning** framework for **facial gesture recognition**, which helps users do oral motor exercise.
- Assisted in implementing **text-to-speech** and **speech-to-text** functions in **Unity** with HuggingFace API and **AWS lambda**.

PROJECTS

[Video Game with Optimization & AI Player](#) 🔗 | *Unity, C#, Firebase*

Worked individually to make a 2D action game with performance optimization and training of AI Player

- Designed and developed game in **Unity**, scripted in **C#**, with **Firebase** for **authentication** and **cloud data save-load**.
- Utilized **state design pattern** to implement game character behaviors, to improve maintainability and scalability of behavior system.
- Implemented **parallel processing** that increases game frame rate by ~90%; and implemented **object pooling** that reduces high frame time occurrence and avoids game stuttering.
- Trained **AI models** that simulate player behavior in playing the game via **reinforcement learning** and **imitation learning**.

[Viral Uprising](#) 🔗 | *Unity, C#*

Worked in a team of 2 to make a 3D shooting & strategic game

- Collaboratively designed and built game in **Unity**, scripted in **C#**.
- Implemented dynamic enemy NPC behavior with aggro system, and movement behavior with A* pathfinding.
- Implemented behavior of strategic buildings deployable by players, and implemented shop and upgrade systems.
- Implemented enemy spawning system that spawns in waves, and implemented game state monitor to determine winning and losing conditions of the game.

[Black White Flip](#) 🔗 | *Unity, C#*

Worked individually to make a 2D puzzle game

- Designed and built game in **Unity**, scripted in **C#**.
- Implemented puzzle logic and game rules, with multiple game mechanics to increase the variety of puzzle levels.
- Implemented game level editing in inspector, to allow rapid and convenient game level creation.

TECHNICAL SKILLS

Language: C#, JavaScript, HTML, CSS, Python, Java, C++, Kotlin, R

Technical Framework: Node.js, Express.js

Technical Tools / Software: Unity, Git, Visual Studio, VS Code, Docker, Firebase, AWS, Android Studio

LANGUAGE

Cantonese - Native

English - Fluent

Mandarin - Fluent